Bank Muamalat X Rakamin

Sales Performance & Transaction Analysis Report (2020-2021)

Data-Driven Insights for PT Sejahtera Bersama

Project-Based Internship : Business Intelligence Analyst
Bank Muamalat X Rakamin Academy

Maulana Zulfikar Aziz

About Me

HELLO, I AM MAULANA ZULFIKAR AZIZ !

A final-year Mathematics undergraduate at Diponegoro University with a strong passion for Data Science and Machine Learning. Experienced in data analytics, predictive modeling, and insight generation through both academic and independent projects.

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Project Goals

DEVELOP AN INTERACTIVE DASHBOARD TO MONITOR SALES PERFORMANCE EFFECTIVELY

Transform raw, complex data into an intuitive visual interface. This allows stakeholders to easily monitor metrics and gain a comprehensive understanding of sales performance without navigating through spreadsheets.

PROVIDE ACTIONABLE INSIGHTS TO SUPPORT STRATEGIC DECISION-MAKING.

Derive meaningful patterns from the visualization to go beyond simple reporting. The ultimate goal is to empower the management team to formulate data-driven strategies for future business growth.

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Dataset Overview

The dataset consists of a single Excel file structured into four relational tables. These tables form the foundation of the analysis, connecting transactional records with product and customer dimensions.

ORDERS

Contains historical transaction records, including order dates, quantities, and foreign keys linking to other tables.

Orders

- OrderID (key)
- Date
- **FK** CustomerID
- **FK** ProdNumber
- Quantity

PRODUCTS

Stores detailed product information such as product names, category, and unit prices.

Products

- ProdNumber (key)
- ProdName
- **FK** Category
- Price

PRODUCT CATEGORY

Classifies products into broader groups (e.g., Robots, eBooks) for higher-level analysis.

ProductCategory

- CategoryID (key)
- CategoryName
- CategoryAbbreviation

CUSTOMERS

Contains customer personal data, such as name, email, phone, address, city, state, and zip.

Customers

- CustomerID (key)
- FirstName
- LastName
- CustomerEmail
- CustomerPhone
- CustomerAddress
- CustomerCity
- CustomerState
- CustomerZip

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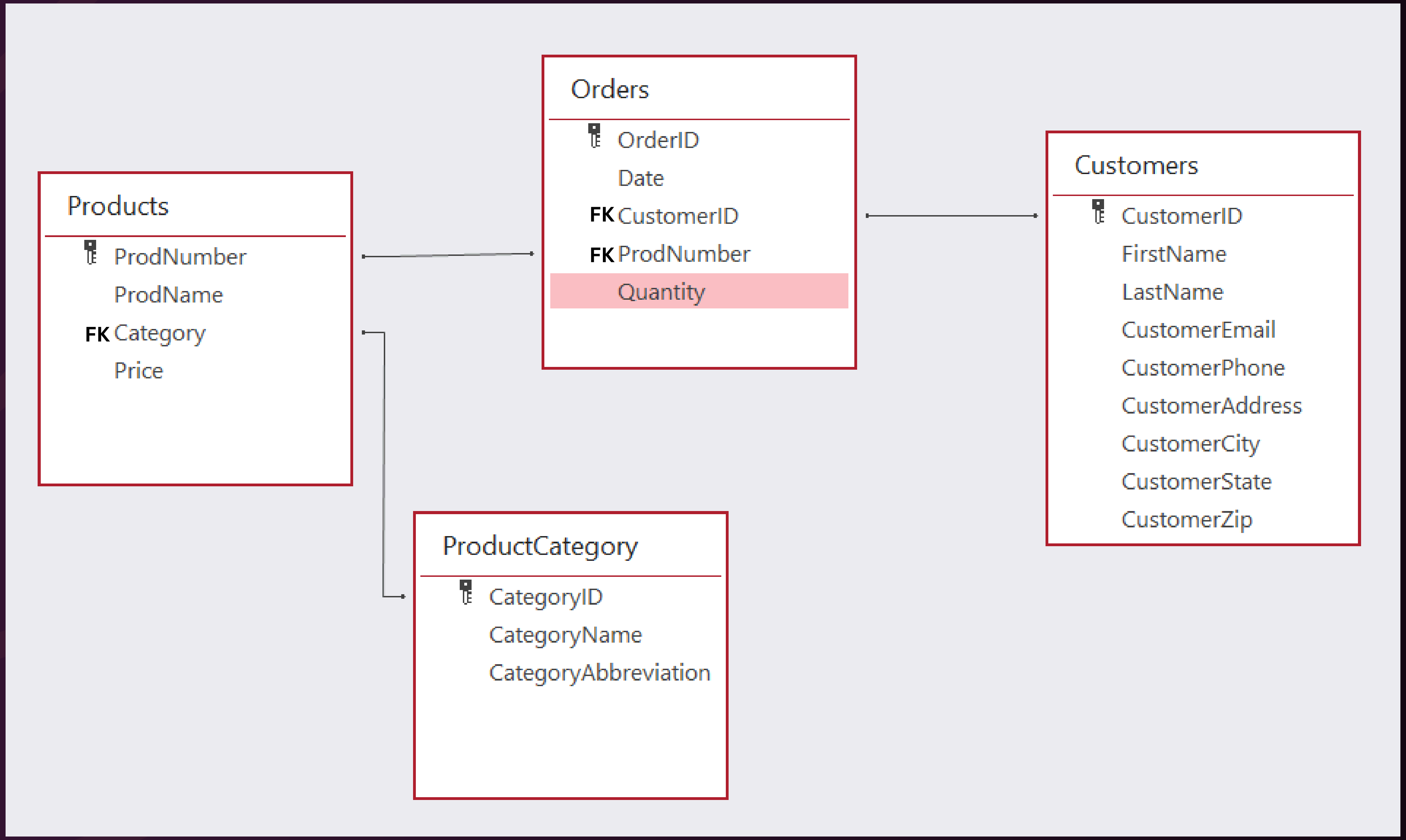
Entity Relationship Diagram (ERD)

The **Orders** table serves as the central repository for transactional data. with **Customers**, **Products**, and **ProductCategory** provide descriptive attributes.

Tables are linked via One-to-Many relationships (e.g., one customer can place many orders).



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Data Preparation : Master Table

Before building the dashboard visualization, we first need to construct a Master Table containing all necessary data points. For this project, I utilized Google BigQuery to import the CSV files and execute the join queries (next slide).

Note on Data Integrity: It is critical to validate the imported data. I discovered that BigQuery initially failed to recognize the decimal values (comma separated) from Excel. To fix this, I performed data cleaning to ensure the revenue figures were accurate.

Data Preparation : Master Table

Master Table Structure

master

- cust_email
- cust_city
- order_date
- order_qty
- product_name
- product_price
- category_name
- total_sales

SQL Query

```sql
CREATE TABLE tugas_akhir_vix.master AS
SELECT
c.CustomerEmail AS cust_email,
c.CustomerCity AS cust_city,
o.Date AS order_date,
o.Quantity AS order_qty,
p.ProdName AS product_name,
p.Price AS product_price,
pc.CategoryName AS category_name,
o.Quantity * p.Price AS total_sales
FROM tugas_akhir_vix.Orders AS o
LEFT JOIN tugas_akhir_vix.Customers AS c ON o.CustomerID = c.CustomerID
LEFT JOIN tugas_akhir_vix.Products AS p ON o.ProdNumber = p.ProdNumber
LEFT JOIN tugas_akhir_vix.ProductCategory AS pc ON p.Category = pc.CategoryID
ORDER BY order_date ASC;
```

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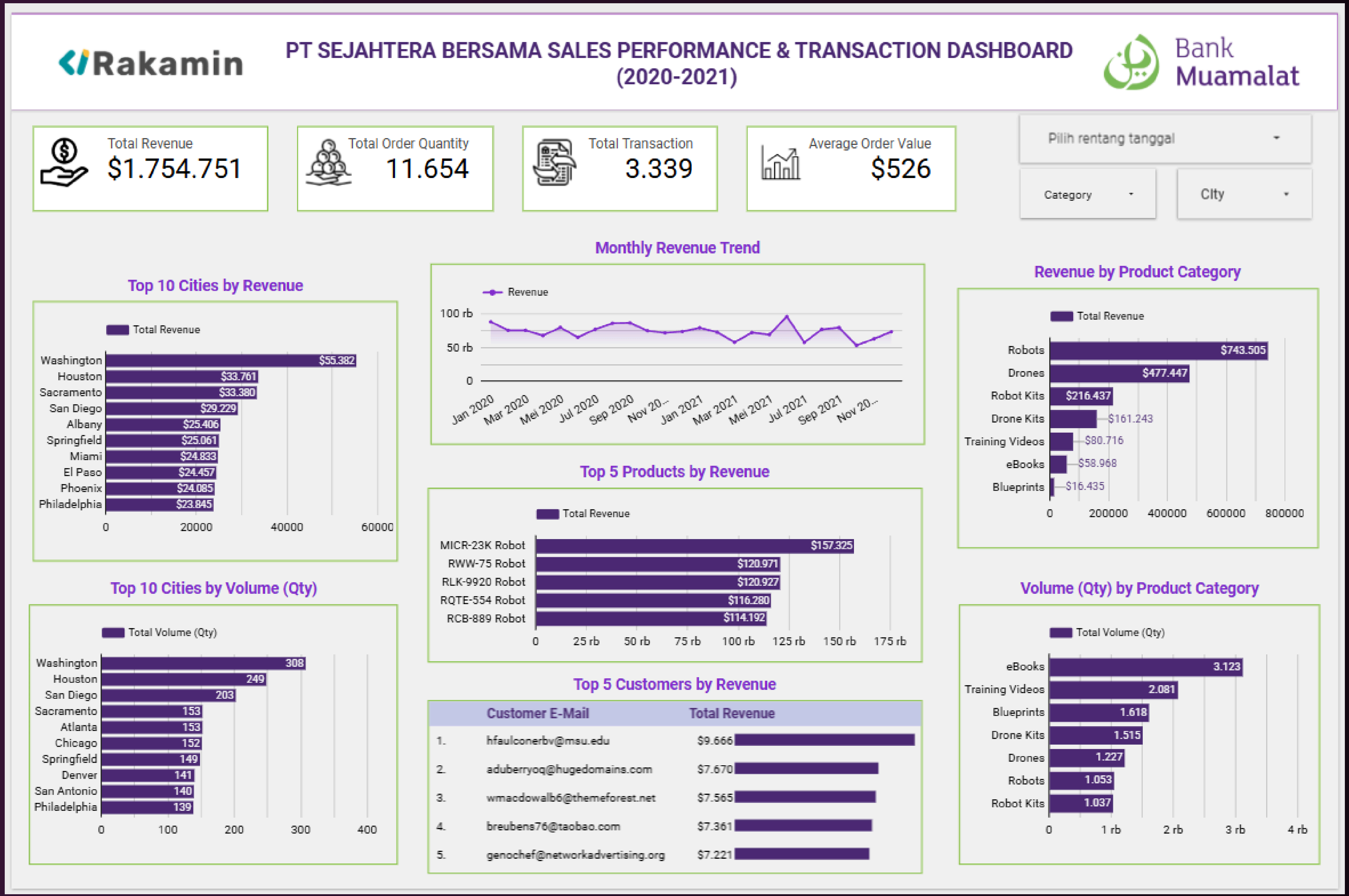
Dashboard



ACCES THE LIVE DASHBOARD HERE

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Insights

- Washington leads in both metrics, generating the highest Total Revenue ($55,382) and the highest Order Volume, indicating a strong market presence in this region.
- The monthly revenue trajectory shows a downward trend over the observed period, signaling a need for strategic intervention to stabilize growth.
- The Top 5 Products list is predominantly occupied by Robots, confirming that this category is the company's primary revenue driver.
- The highest-contributing customer is customer with e-mail hfaulconerbv@msu.edu, generating significant revenue for the company.
- Aligned with product performance, 'Robots' and 'Drones' are the most profitable categories. These two segments significantly outperform all others in terms of monetary value.
- In contrast to revenue, 'eBooks' and 'Training Videos' rank highest in Sales Volume. This indicates high demand for affordable items, though they contribute less to the total bottom line compared to tech hardware.

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Business Strategies

REGIONAL STRATEGY: DOMINATE KEY MARKETS

Strengthen market leadership in Washington by intensifying localized marketing campaigns. We should capitalize on the high transaction volume in this region to maximize revenue potential and set it as a benchmark for other cities.

GROWTH STRATEGY: REVERSE THE DOWNWARD TREND

Counteract the downward revenue trend by implementing a dual-strategy: aggressive customer acquisition through targeted advertising and a rigorous customer feedback loop to identify and resolve churn drivers.

PRODUCT FOCUS: HIGH-VALUE & CROSS-SELLING

Shift marketing budget allocation towards 'High-Margin' products (Robots & Drones) to boost profitability. Simultaneously, utilize 'High-Volume' products (eBooks) as an entry point to cross-sell expensive hardware (e.g., Bundling Strategy: 'Buy a Robot, Get a Training Video Free').

CUSTOMER RELATIONSHIP MANAGEMENT (CRM)

Implement a Key Account Management approach for top-tier clients. Provide exclusive perks or personalized support to ensure retention and incentivize them to become brand advocates through referrals.

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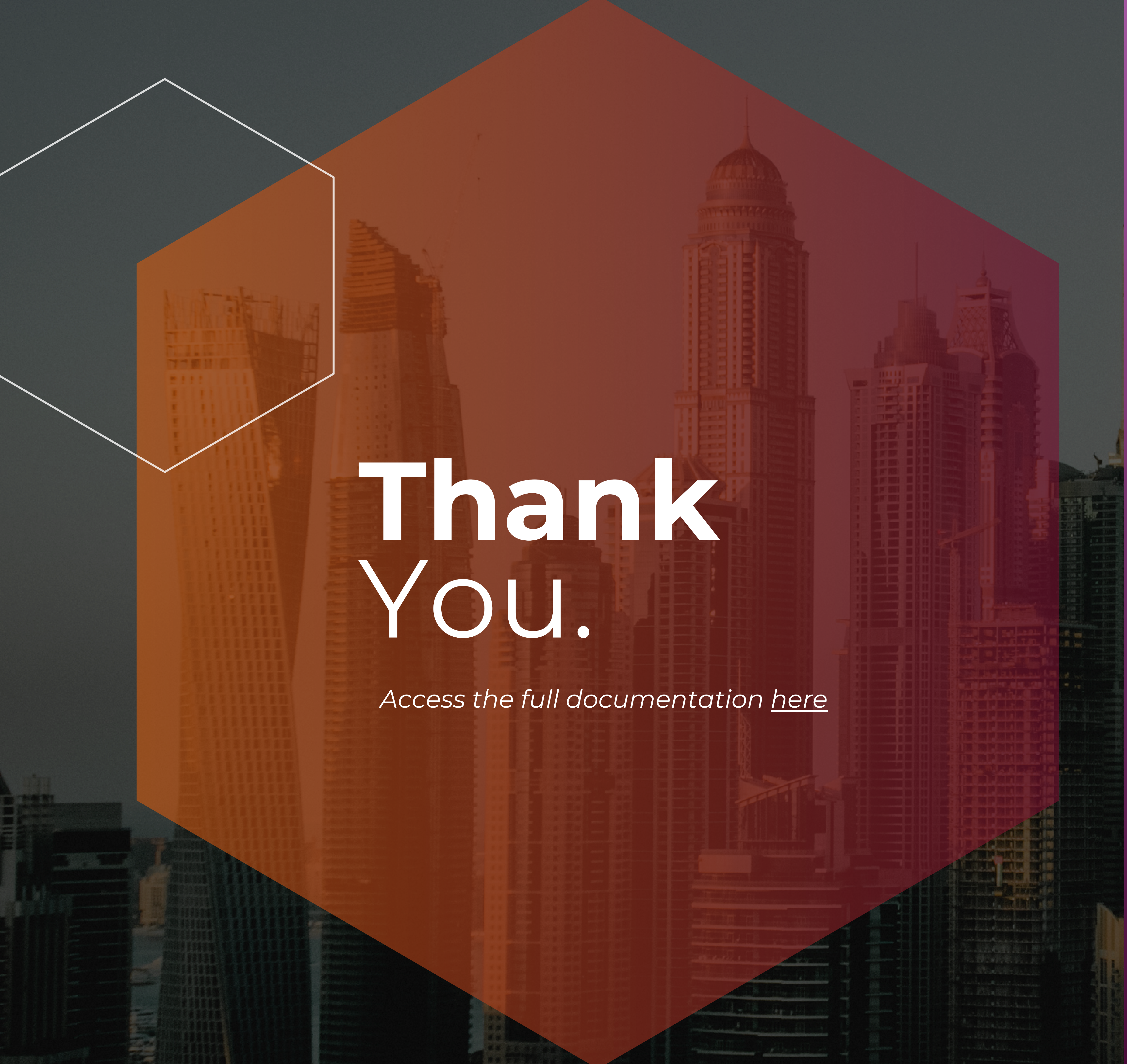

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Thank You.

Access the full documentation here

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